

Only the Constants Survive: Why Invariance-Based Demarcation Has Nothing to Hold at the Register of Semantic Decomposition

An admissibility-collapse result for the invariance programme in logic, developed under Trisductive verification discipline

Mohammad F. Islam¹

¹Independent researcher, United States. Correspondence: islammm@alumni.iu.edu. Preprint. Source and reproduction materials at the repository named in the References.

Since Mautner in 1946 the invariance programme has demarcated logical notions as those preserved by a group acting on the domain of discourse, a strategy refined through Tarski, Sher, McGee, Feferman, Casanovas, and Bonnay into the modern invariance criteria for logicity. The programme's finiteness underwriting comes from classical invariant theory: for a linearly reductive group acting linearly on a finite-dimensional space the invariant ring is finitely generated, and for the classical groups its generators are named explicitly. This paper identifies a register at which that machinery cannot run, and shows the failure is structural rather than technical. The invariance criterion succeeds on the domain register because permutations of a domain form a group by construction. It fails on the register of semantic decomposition, where the operative transformation is not permutation but abstraction, the discarding of a constituent, which is idempotent and carries no inverse. The transformation set is therefore a monoid once the empty abstraction is adjoined, group-hood fails before reductivity is ever tested, and neither the Hilbert finiteness theorem nor the Weyl explicit generators apply. We give the obstruction in exact form, an operator with singular values one, one, zero whose distance from left-invertibility is bounded below by exactly one in spectral norm and attained there. Admissibility, that every constituent of a claim is abstractable, then forces a stronger conclusion than the failure of a finiteness theorem. Invariance under each of the three coordinate abstractions forces every invariant function to be constant on the whole space in three substitutions, equivalently because the semigroup they generate contains an annihilator: finite generation holds vacuously, generated by the constant one, while the separating power collapses absolutely, no invariant distinguishing any two points whatever. The criterion at this register does not lack its underwriting theorem. It demarcates nothing. We show the consequence for any pipeline composing a semantic decomposition with a decidable arithmetic verdict, namely that the non-effectiveness Church and Tarski require of a truth-tracking procedure must reside on the decomposition side since the remaining stages are finite and decidable, and we state the one hypothesis on which the collapse turns, which is not closure under composition, that being automatic, but admissibility: that every constituent is abstractable.

KEYWORDS invariance criteria for logicity; Hilbert's fourteenth problem; finite generation; separating invariants; monoid actions; semantic decomposition; effective procedures; Trisduction

Background and rationale

Klein's 1872 programme fixed a definition of geometry that has outlived every particular geometry it was built to organize: the

propositions of a geometry are exactly the relations invariant under its acting group, and the hierarchy of geometries is the hierarchy of those groups¹. The move that concerns us here was made seventy-four years later, when Mautner extended the programme out of geometry and into logic under a title that states the thesis without ornament, logic as

invariant-theory². Tarski gave the idea its canonical form in a 1966 lecture published two decades afterward, proposing that logical notions are those invariant under arbitrary permutations of the domain of discourse³. Sher developed it into a full demarcation criterion⁴, McGee characterized the resulting class in terms of definability in an infinitary language⁵, and Feferman, Casanovas, and Bonnay refined, criticized, and generalized the criterion through homomorphism invariance and potential isomorphism^{6,7,8,9}.

What the programme buys, and why. A criterion of this shape is not a heuristic. It is a demarcation with a finiteness theorem behind it. When a group acts linearly on a finite-dimensional space, one may ask whether the invariant functions form a finitely generated algebra, and for a linearly reductive group Hilbert answered yes in 1890^{11,12}, with Noether supplying the finite-group case and a degree bound¹³, Weyl naming the generators explicitly for the classical groups¹⁴, and Haboush closing the positive-characteristic case¹⁶. This is the load the programme silently carries. A criterion that says the admissible notions are the invariants is only a usable criterion if the invariants can be enumerated, and it is Hilbert who guarantees they can.

The guarantee is conditional and the condition is not decorative. Nagata gave a linear action of a non-reductive group whose invariant ring is not finitely generated¹⁵, and the negative answer to Hilbert's fourteenth problem has since been reproduced in progressively smaller dimensions by Roberts and by Daigle and Freudenburg^{18,19}. Popov strengthened the picture: for every non-reductive algebraic group there exists an affine variety on which the invariant ring fails to be finitely generated³⁵. Finite generation is therefore a privilege purchased by reductivity, never a general fact about invariants, and any register that hopes to inherit the invariance criterion must first establish that it has a group and that the group is reductive.

The barrier this paper isolates. The programme has always run on a single register, the domain. Permutations of a domain form a group by construction, so the first hypothesis is met without argument and the second is met because the symmetric group is reductive. Every result from Mautner through Bonnay lives there. What has not been asked is whether the same criterion transports to a different register, the one on which a proposition is decomposed into constituents before any interpretation is fixed. This is not the register of what the domain's objects are; it is the register of how a claim is carved.

The barrier is that this second register supplies no group. Its operative transformation is abstraction, the discarding of a constituent, and abstraction is not a permutation. Permutation rearranges without loss and admits an inverse. Abstraction removes, and what is removed cannot be reconstructed from what remains: two distinct claims that differ only in the discarded constituent have the same image, so the map is not injective and no inverse exists on any subset that contains both. The distinction is already visible in applied work: in the corpus-augmentation literature it is a working rule that a transformation which creates ambiguity is often semantically invariant while one that resolves an ambiguity is not, which is exactly the asymmetry between a lossy map and its would-be inverse, observed empirically and left untheorized⁴².

Why the barrier is structural, not computational. It is worth being precise about the kind of failure at issue, because the two kinds are routinely confused. A computational barrier is one where the object exists and we cannot yet compute it: the invariant ring is finitely generated but no algorithm produces the generators in feasible time. A structural barrier is one where the object does not exist because a hypothesis of the theorem that would produce it is unmet. Nagata's counterexample is structural in this sense with respect to finite generation, and the failure identified here is structural one level earlier, at group-hood rather than at reductivity.

The distinction matters because it fixes what would count as progress. If the barrier were computational, better algorithms would dissolve it, and the substantial machinery of computational invariant theory would be the place to look¹⁷. If it is structural, no algorithm helps, and the only routes forward are to supply a group where none was found, to weaken the closure demanded until something available suffices, or to accept the negative result and read off its consequences. This paper argues for the third route while stating precisely what the first two would require.

The domain of validity, mapped. The standard programme's domain of validity is exactly the register where a group acts on a set of objects and the invariants sought are properties of relations over that set. Inside that domain the criterion is correct, the finiteness result applies, and the debates in the literature are about which group is the right one, not about whether a group is available. Outside that domain, on the register of decomposition, the criterion has no acting group to read invariants from and therefore does not fail to hold so much as fail to parse: a demand for the invariants of a monoid action under a theorem that quantifies over reductive groups is not a false statement but an ill-typed one.

One further hypothesis that is usually skipped. Even where finite generation fails, invariant theory does not collapse into nothing, and the honest statement of the negative result must account for this. Nagata himself proved that invariant rings are at least rings of regular functions on a quasi-affine variety, and Derksen and Kemper proved that a finite separating set always exists: a finite collection of invariants that distinguishes any two points distinguished by any invariant at all¹⁷. Separation is weaker than generation, but it is a genuine closure and it survives the failure of Hilbert's fourteenth. Any claim that a register admits no closure whatever must therefore clear separation as well as generation. We clear both below, and in opposite directions: generation holds vacuously and separation degenerates to nothing.

The shape of the result. What follows is a negative result with a named mechanism and a stated ceiling. The mechanism is the non-invertibility of abstraction, given in exact linear-algebraic form and executed. The ceiling is set by one hypothesis, that every constituent of a claim is abstractable, stated where it is used rather than treated as a formality; granted it, the invariant ring is not merely unavailable but computed, and found trivial. The proposed result develops the obstruction and its consequence for effective procedures. The falsifiable commitments state four conditions under which the result would be refuted.

Brief literature review

Five bodies of work bear directly on the claim, and each is given its due before its structural limitation is named.

The invariance programme in logic. From Mautner² through Tarski³, Sher⁴, McGee⁵, Feferman^{6,7}, Casanovas⁹, and Bonnay⁸, the programme demarcates logical notions by invariance under a transformation group on the domain. Its achievements are real: it converts a philosophical question about the nature of logic into a mathematical one with definite answers, and McGee's characterization in terms of infinitary definability shows the resulting class is not arbitrary. Its known limitations are internal to the register. The criterion overgenerates, admitting operations that are intuitively non-logical, which motivated Feferman's and Bonnay's refinements; and it undergenerates, since permutation invariance yields notions sensitive essentially only to cardinality, which reduces logic to a logic of isomorphism type and drops modal and intensional phenomena¹⁰. What no participant asks, because the register never raises it, is whether the criterion's underwriting theorem transports beyond a domain group. The programme is not wrong here; it is silent, and this paper reads the silence rather than contradicting the speech.

Classical invariant theory. Hilbert's finiteness theorem^{11,12} with Weyl's explicit generators¹⁴ is what makes the invariance criterion a criterion rather than a definition. The limitation is the reductivity hypothesis, made unmistakable by Nagata¹⁵ and by the smaller counterexamples that followed^{18,19}, and softened but not removed by the separating-set theorem¹⁷. The literature is precise about which groups it covers. It does not treat non-group transformation sets at all, and there is no analogue of Hilbert's theorem for a general monoid acting on a structure that is not an algebraic variety.

Geometric deep learning. The most visible recent transport of the Erlangen programme is the geometric deep learning blueprint, which derives architectures from the symmetry group of the data domain, grids under translation, sets and graphs under permutation, spheres under rotation^{29,30,31}. It is the strongest available evidence that Klein's move generalizes far outside geometry, and it is honest about its scope: the blueprint applies where a symmetry group is identified. Language enters that framework through sequence and permutation structure on tokens, not through a group on semantic decompositions, so the blueprint does not address the register at issue and does not claim to.

Geometric models of paraphrase. There is active work modelling paraphrase as a continuous transformation in embedding space, including a recent framework that treats paraphrasing as a structured affine transformation inspired by local Lie group actions and decomposes paraphrase transitions into rotation, deformation, and translation³⁹. This is the closest live counterparty to the present claim and must be met rather than dismissed. Two observations resolve the tension without deflating the work. First, the object is an embedding, a finite-dimensional real vector space imposed by a model, and a group can always be made to act on such a space; the question is whether the action is a faithful representation of the semantic operation or an approximation chosen for tractability, and the

reported area under curve of roughly 0.77 on a noisy corpus is consistent with approximation. Second, and decisively, an affine or Lie-group model of paraphrase is a model of meaning-preserving transformation. Abstraction is not meaning-preserving; it is meaning-reducing, which is why it has no inverse. A group model of the invertible sub-part of the transformation set is compatible with the non-group character of the whole.

Compositional semantics and the formalization gap.

Montague's programme and its successors give compositional interpretation procedures for fragments of natural language²⁸, and the tradition is aware that the hard step is the passage from an informal claim to its formal rendering rather than any step after it. What the tradition does not do is locate that hardness by exclusion, showing that it cannot reside in the later stages because those stages are decidable. That argument is given under the proposed result and is, so far as we have found, the paper's one genuinely unoccupied position.

The common structural error. Every approach above shares one assumption, held so uniformly that it is rarely stated: that where invariance analysis is wanted, a group will be available, and the only live questions are which group and how large the resulting class is. The assumption is correct on the domain register and unexamined off it. The proposed result shows that it fails at the first register one moves to, and that the failure is diagnosable in three lines of linear algebra.

Methodology

The claim advanced here is negative, and negative claims are easy to state loosely and hard to state at their true strength. We therefore fix in advance the three conditions any component of the argument must satisfy, and hold every later claim to them.

Condition one, formal derivation. Each load-bearing step must follow from a stated classical theorem or from elementary linear algebra with the derivation given in the text, not deferred to a citation that the reader must reconstruct. Where a step is conditional on a hypothesis, the hypothesis is named at the point of use. Where a step is a placement rather than a derivation, it is labelled a placement and carries no theorem's warrant.

Condition two, computational signature. Every quantitative assertion must have an executable witness at a declared seed, so that a reader with a standard scientific stack can reproduce the number rather than accept it. The battery reported under the proposed result was executed at seed 20260622 in double precision. Assertions that admit an exact analytic bound carry the bound and use the execution only as confirmation, never as substitute; assertions that are sampled are labelled sampled.

Condition three, invariance under reframing. The argument must not depend on the particular vocabulary in which the decomposition register is described. If the result held only under one description of what a decomposition is, it would be an artefact of that description. We therefore state the obstruction at the level of the operator rather than the level of the linguistic theory, so that any

account on which some constituent is discarded instantiates it, whatever the account calls the constituents.

Independence verifiability. The three decisive checks in this paper are a spectral bound, an idempotence identity, and a twenty-seven-point orbit count, the first two computable from a three-by-three matrix by any linear algebra library in any language and the third checkable by hand, together with a citation check against the invariance and invariant-theory literatures, executable in any bibliographic database. None of these requires access to the author, to a specific implementation, or to any private dataset. This matters more for a negative result than for a positive one: a claimed impossibility that only its author can check is indistinguishable from a claimed impossibility that is false, and the safeguard against that is that every step here is checkable by parties who share nothing with the author but a computer and a library.

What the method does not supply. The method verifies and audits; it produces no mathematics. Every theorem invoked is classical and cited, and the contribution is the arrangement, the identification of a register at which known hypotheses fail. No new invariant theory is claimed, and the paper's conclusions rest on the cited results alone.

The proposed result

The mechanism is stated first in plain terms and then in exact ones. On the domain register, the transformation that defines the invariance criterion is a permutation, which is invertible, so the transformations form a group, so Hilbert's theorem is available and the criterion has a finite catalogue behind it. On the decomposition register, the transformation that defines the analogous question is an abstraction, which discards a constituent and is therefore not invertible, so the transformations form a monoid, so no closure theorem in the resident machinery applies. That is the first half. The second is that granting admissibility, that every constituent is abstractable, the invariant ring is not merely unreachable by that machinery but computable and trivial. The interest of both lies in how little it takes to establish them and how much they constrain.

The obstruction in exact form

Let a complete claim carry three constituents. Nothing here depends on the number being three or on any particular theory of what the constituents are; the argument runs identically for any finite count above one, and three is used because it is the count on which the deletion discipline of the framework under which this paper was developed operates, and because it makes the matrices small enough to print.

Represent the constituent occupancy of a claim as a vector in a three-dimensional real space, one coordinate per constituent, and represent the operation of discarding the third constituent as the linear map

$$D = \text{diag}(1, 1, 0).$$

Three facts follow immediately and each is exact.

D is not injective. The vectors $(2, 5, 7)$ and $(2, 5, -3)$ are distinct and have the same image under D , so two claims differing only in the

discarded constituent are indistinguishable after abstraction. The singular values of D are 1, 1, 0 and its rank is 2. Nothing is recoverable across the kernel, which is the formal content of the ordinary observation that a summary does not determine what it summarized.

D has no left inverse, with the deficiency exactly one. Suppose A were a left inverse, so that $AD = I$. Let v be a unit vector spanning $\ker D$. Then $(AD - I)v = ADv - v = -v$, so $\|(AD - I)v\| = 1$, and since the spectral norm is the supremum of $\|Mx\|$ over unit x ,

$$\|AD - I\|_2 \geq 1 \text{ for every } A \text{ whatever.}$$

The bound is sharp and attained, for instance at $A = I$, where $AD - I = \text{diag}(0, 0, -1)$ has singular values 1, 0, 0 and spectral norm exactly 1; $A = D$, $A = D^+$, and $A = 0$ attain it likewise. Attainment is irrelevant to the conclusion, since what matters is that the distance is bounded below by 1 for every A whatever, so no A satisfies $AD = I$. The equality set is characterized: writing the columns of A as a_1, a_2, a_3 , equality holds exactly when $a_1 - e_1$ and $a_2 - e_2$ lie in the span of e_1 and e_2 with the three-by-two block they form of spectral norm at most 1. This is elementary, and that is the point: the obstruction requires no deep theorem, which is why it has been available for inspection all along and why its consequence rather than its proof is the contribution.

D is idempotent. $DD = D$ exactly. A cut cannot be re-cut. Together with the previous fact this fixes the algebraic type: the transformation set generated by abstractions is closed under composition and contains non-invertible elements, so it is not a group. It is a monoid once the empty abstraction is adjoined, which is a declaration rather than an observation, since the generated semigroup does not itself contain the identity; the enumeration is given below and the point does not depend on the convention.

Box 1 | Executed battery LOG-CHK, seed 20260622, double precision

Nine checks. Where an exact statement exists it is the claim and the execution is confirmation; where a figure is local execution evidence it is labelled so. No sampled figure is offered as evidence for a proved bound.

LOG-CHK.1, the domain register has a group. Twenty thousand sampled orthogonal three-by-three matrices, maximum deviation of QQ^T from the identity 1.443×10^{-15} , closure under inverse confirmed to machine precision.

LOG-CHK.2, the group orbit is separated there. Two configurations forced to identical Gram matrices, agreement 2.842×10^{-14} , connected by an orthogonal map with maximum deviation of OO^T from the identity 4.441×10^{-16} . The Gram entries separate the orbits: equal Gram matrices imply relation by an element of the acting group. This is the separating property and not finite generation. Finite generation for the orthogonal group is Weyl's theorem, cited and not exhibited here, and the distinction is the one the commitments turn on.

LOG-CHK.3, the decomposition operator is not injective. Singular values 1, 1, 0; rank 2; the inputs (2, 5, 7) and (2, 5, -3) carry to a common image at separation exactly 0.

LOG-CHK.4, the floor is proved and attained. Exhibited minimizers $A = I$, $A = D$, $A = D^+$, and $A = 0$, each returning $\|AD - I\|_2 = 1.000000000000$ exactly, the first three with singular values 1, 0, 0 and the last with 1, 1, 1. The floor is the analytic claim of the text; these are witnesses to its sharpness. A random sweep is not reported, because its minimum tracks the sampler's scale parameter rather than any property of D , approaching 1 from above as the scale shrinks and never breaching a proved bound.

LOG-CHK.5, idempotence. $|DD - D| = 0$ exactly.

LOG-CHK.6, the generated semigroup, enumerated exhaustively. The three coordinate abstractions generate seven elements, every zero-one diagonal except the identity. The identity is provably absent, since each generator has rank 2 and rank does not increase under composition. The annihilator, the zero matrix, is present, obtained as the product of all three distinct generators in any order; a product that repeats a generator does not annihilate, since the generators are diagonal and therefore commute and each is idempotent, so a repeated factor adds nothing. Six of the twenty-seven ordered triples annihilate, exactly those using all three.

LOG-CHK.7, the invariant ring collapses. The annihilator sends every x to 0, so any f invariant under it satisfies $f(x) = f(0)$ for every x and the invariant ring is the constants. This is an identity and carries no sampled figure.

LOG-CHK.8, the finitary core, hand-checkable. Occupancy vectors over three values per coordinate, twenty-seven points, orbits computed by union-find under the three coordinate abstractions. Result: one orbit class, so the invariant ring is one-dimensional, matching the seven-element semigroup which also returns one class. Admissibility denied: under D_3 alone, nine classes, the functions of the two surviving coordinates; under D_2 and D_3 , three classes. The

collapse and its denial are both exhibited on a model small enough to check by hand.

LOG-CHK.9, graded abstraction, the open case computed, in two halves that carry different weight. A graded abstraction attenuates a coordinate rather than zeroing it and is therefore not idempotent, so it generates a different transformation set. On a sixty-four-point model, values zero through three per coordinate with the graded map halving a coordinate toward zero, three graded maps leave one orbit class, matching the three sharp ones exactly; two leave four and one leaves sixteen, against four and sixteen for the corresponding sharp restrictions. That half is weaker than it reads and the reason is worth printing: on integer values the halving map reaches zero in finitely many steps, from three by way of one under flooring and from three by way of two and one under rounding, so the graded semigroup on this model contains the annihilator outright and the collapse holds there for the ordinary reason, with continuity playing no part.

Over a real coordinate space attenuation never reaches zero in finitely many steps, and that is where the real content sits. Invariance under attenuation by a factor L in the open unit interval gives $f(x_1, x_2, L^n x_3) = f(x)$ for every n ; the sequence $L^n x_3$ converges to zero, and for nonzero x_3 without attaining it; and continuity gives $f(x_1, x_2, x_3) = f(x_1, x_2, 0)$, so the sharp map is recovered in the limit and the collapse follows. Continuity is doing the work, and dropping it exhibits the escape rather than merely naming it: the indicator of a nonzero third coordinate is invariant under all three graded maps, since attenuation by a nonzero factor preserves nonzeroness, and it is not constant, taking the value one at any point with nonzero third coordinate and zero at any point without. This is an identity and carries no sampled figure. The same indicator is not invariant under the sharp third abstraction, which sends a nonzero coordinate to zero. The collapse is therefore robust under grading for continuous invariants, and the escape it offers is exactly the discontinuous one just exhibited.

Warrant: theorem-grade on the spectral floor, the idempotence, the exhaustive enumeration, the finite-model orbit counts, the graded-abstraction counts and continuity argument, and the collapse; engineering-grade on the two sampled deviations in checks one and two, which are local execution evidence. Checks three through nine are exact and require no seed.

The collapse, and the one hypothesis it turns on

The obstruction so far is the absence of a group. Admissibility forces something sharper, and it answers the question the finiteness machinery was being asked in the first place. Closure under composition is entailed by the invariance relation and is not an additional hypothesis, as the subsection below shows.

The three coordinate abstractions generate a seven-element semigroup, LOG-CHK.6, and its elements are every zero-one diagonal matrix except the identity. Two facts about that set are decisive. The identity is absent, since rank does not increase under composition, so the monoid claim holds only after the empty abstraction is adjoined by declaration; the convention is harmless and the substantive point survives without it, because a set containing non-invertible elements is not a group under any adjunction. And the

annihilator is present, since composing abstractions of all three coordinates discards everything.

Admissibility settles the invariant ring immediately, and in three substitutions. Let f be invariant under each of D_1, D_2, D_3 . Then $f(x_1, x_2, x_3) = f(0, x_2, x_3) = f(0, 0, x_3) = f(0, 0, 0)$, each step an application of one generator at one point, with no composition anywhere. So f is constant, and

the invariant ring is $\mathbb{R}$.

Equivalently, and as a corollary rather than a step, the semigroup those generators generate contains the annihilator, and invariance under the annihilator gives the same conclusion in one line. The two readings are the same fact from either end, and the substitution chain is the minimal one.

Both closure questions therefore answer, and they answer in opposite directions. Finite generation holds, vacuously, the ring being generated by the constant one. The separating power collapses absolutely, no invariant distinguishing any two distinct points; separating sets exist and separate nothing, as the commitment below states exactly. The register does not suffer a weaker closure than the domain register enjoys. It has an invariant theory that is trivial and a demarcation criterion that demarcates nothing, which is a stronger negative than the failure of a finiteness theorem and a different one.

The hypothesis, stated so it can be denied. The collapse does not turn on closure under composition, and it is worth saying why, since closure is the natural place to look and an earlier draft of this paper looked there. The set of maps under which a given function is invariant is closed under composition automatically: if $f \circ g = f$ and $f \circ h = f$, then $f \circ (g \circ h) = (f \circ g) \circ h = f \circ h = f$. A discipline admitting single abstractions but barring their iteration therefore bars nothing, because the invariance conditions compose whether or not the maps are declared to.

Run directly, the collapse needs only the three generators applied at three points, and no composition occurs anywhere in it. Invariance under D_1 at x gives $f(x_1, x_2, x_3) = f(0, x_2, x_3)$. Invariance under D_2 at the point $(0, x_2, x_3)$ gives $f(0, x_2, x_3) = f(0, 0, x_3)$. Invariance under D_3 at $(0, 0, x_3)$ gives $f(0, 0, x_3) = f(0, 0, 0)$. The annihilator is entailed rather than assumed, and it was never doing the work.

What the collapse does turn on is admissibility: that every constituent is abstractable, so that all three generators are in the transformation set. That hypothesis is genuinely deniable and denying it restores a non-trivial ring. Invariance under D_3 alone leaves the functions of the two surviving coordinates, nine orbit classes of twenty-seven on the finite model of LOG-CHK.8 rather than one; invariance under D_2 and D_3 leaves three. The reader who wants to resist should resist at admissibility, and not at closure, which is entailed, nor at the algebra, which is three substitutions.

Where admissibility comes from, and the domain it ranges over. The deletion discipline supplies admissibility directly, and this is worth stating plainly, because in an earlier draft the appendix appeared to be an objection and it is in fact the premise. The

discipline establishes the constituent count by deleting each constituent in turn and observing that the residue is no longer a complete claim. Deleting each in turn is precisely the admission of all three generators, which is the hypothesis the collapse needs.

One honest gap travels with that. The appendix presents deletion as a method for counting constituents, not as a claim that every constituent may be abstracted in the course of ordinary use, and those are different claims: the first supplies admissibility for the counting operation and the second supplies it for the register generally. The collapse needs the second, and the first is evidence for it rather than a proof of it. A register in which some constituent is countable by deletion but not abstractable in use would escape, and we do not know of one.

The domain the invariants range over is worth naming in the same place, since a first abstraction already leaves the class of complete claims. The transformation semigroup acts on occupancy vectors in a three-dimensional real space, not on complete claims. The first abstraction already leaves the class of complete claims, by the appendix's own observation, so the register on which invariants are sought contains non-claims from the first step onward and composition on it is unproblematic. What the collapse establishes is therefore a statement about the occupancy register: the invariance criterion, run there, demarcates nothing. It is not a statement about a category of complete claims closed under abstraction, because no such category exists, which is the appendix's point and not a difficulty for it. A reader who prefers the narrower reading may restrict the result to the vector register explicitly, at no cost to anything the abstract claims.

What the algebra is, and what the contribution is not. The collapse itself is a standard fact wearing an unfamiliar name. An element that absorbs every product is called a zero element in semigroup theory, and any transformation semigroup containing one has constant invariants, immediately and for the reason given above. Nothing in that observation is new and nothing here claims it is. The contribution is the identification: that the transformation set of a register in which every constituent is abstractable has only constant invariants, equivalently that the semigroup it generates contains a zero, the two being the same fact seen from either end and the first being the minimal statement. A referee who reaches for the semigroup literature will find the algebra there^{40,41} and will not find the identification, and that division is the correct one to state in advance.

Where the result sits, and what that earns. The collapse holds over any set carrying a distinguished point and the three coordinate abstractions, and the set may be taken finite. On the twenty-seven-point occupancy model of LOG-CHK.8 the three generators leave a single orbit class, so the invariant ring is one-dimensional, and the whole verification is a union-find over twenty-seven points that a reader runs by hand in a few minutes. That finitary statement is the collapse; the presentation over a real coordinate space is a convenience.

Stated finitarily the result involves no coding of syntax, no provability predicate, no diagonal construction, no metatheory, and no infinitary comprehension, so the incompleteness and undefinability results have

no jurisdiction over it, not because it evades them but because it never enters the arena they presuppose.

That location earns three things and it is worth being exact, because the rest of the paper does not share them. The collapse inherits no determinacy premise about the natural numbers forming a definite totality, so nothing in it is capped at the warrant of that posit. It inherits no soundness or consistency assumption, since no formal system is quoted. And it is checkable by hand in finite time by a reader who accepts nothing beyond finite set closure and the definition of an invariant.

Two components are excluded from that claim and are declared here rather than absorbed. The presentation over a real coordinate space quantifies over real-valued functions on that space, which is a commitment of a different kind from the determinacy premise and not obviously lighter or heavier than it; we make no ranking, and the finitary statement above is what carries the three items. And the spectral floor is real-analytic, a supremum over the unit sphere, so it is not hand-checkable in the sense just claimed and is not claimed for. Under the warrant-typing rule a composite inherits its weakest component, which is exactly why the collapse and the floor are stated as separate claims about separate objects.

The residence argument does not sit there and the paper should not be read as if it did. That argument invokes Church, Turing, and Tarski as premises, so it lives inside the stratum those theorems measure and inherits everything that stratum inherits: it is conditional where the collapse is unconditional, and an attack on the undecidability leg moves it while leaving the collapse untouched. The two halves therefore stand or fall separately, which is the correct architecture for a paper carrying a negative result and is stated here so that a reader who rejects the corollary does not take the theorem down with it.

Why the failure precedes reductivity

It is worth being exact about where in the chain of hypotheses the failure occurs, because a failure at a later link would be a different and weaker result. Hilbert's theorem requires a linearly reductive group acting linearly on a finite-dimensional space. Nagata's counterexample shows that dropping reductivity can cost finite generation. The decomposition register never reaches that question. It fails at group-hood, one link earlier, and this has a consequence that is easy to miss: because the failure is earlier, the substantial theory built to handle non-reductive actions, including the separating-set result¹⁷, is equally unavailable, since that theory too quantifies over groups acting on varieties. The register does not obtain a weaker closure. It obtains no closure from this machinery at all, and the orbit count above shows separately that the closure it does have is trivial. These are two findings, and the second does not depend on the first.

The one candidate that would supply a group

The natural repair is to propose synonymy as the acting relation. If meaning-preserving re-description were invertible, paraphrase would form a group and its invariants would be meanings, and the criterion

would transport after all. Two observations block the repair in the setting at hand.

The first is that synonymy and abstraction are different operations and the register requires both. A decomposition discipline that admits only meaning-preserving moves cannot decompose, since decomposition is precisely the isolation of a constituent by setting others aside. The invertible sub-part of the transformation set may well form a group; the set as a whole does not, and the closure theorem quantifies over the whole.

The second is that in the specific discipline motivating this analysis, synonym transport is forbidden by an independent requirement. The decomposition procedure demands that the vocabularies populating the constituents be pairwise disjoint and rejects any decomposition in which they collide, precisely so that the constituents can be varied independently. That requirement bars the transport of a term across constituents, which is the operation a synonymy group would need to perform. The first operation destroys invertibility and the second forbids the transformation that would restore it, and both are motivated on grounds having nothing to do with the present result.

The consequence for effective procedures

The result would be a curiosity if it stopped at the algebra. It does not, because it constrains where the non-effectiveness of a whole class of pipelines must sit.

Consider any procedure that takes a claim, decomposes it, screens the decomposition through a finite roster of failure tests, and computes a verdict by a fixed arithmetic operation on the resulting numerical rows. The screening stage has a finite enumerated roster and is decidable. The verdict stage is a determinant computation at finite precision, resolving to one of finitely many tokens, and is decidable. If the decomposition stage were also effectively closed, in the sense that a finite catalogue of admissible decompositions with decidable membership were available, then the composite would be an effective procedure whose verdicts are claimed to coincide with truth over an arithmetic domain. Church's and Turing's results, and Tarski's undefinability theorem, forbid exactly that^{21,22,24}. They do not forbid effective procedures over arithmetic domains as such, and a proof checker is a standing counterexample to any reading on which they would; what is forbidden is a procedure that decides truth on a class where truth is undecidable. The premise is stated here because the argument is invalid without it.

The non-effectiveness therefore has to reside somewhere in the pipeline, and it cannot reside in the last two stages, which are finite and decidable by construction. It resides in the first, and in the supply of any external certificate the procedure requires. This is a placement, not a theorem about semantics, and it is stated at that grade. Its force is that it converts a familiar practitioner's complaint, that formalization is the hard part, into a constraint with a location: the hardness is not distributed across the pipeline and not reducible by improving the later stages, because the later stages have no room to carry it.

What the standard result becomes. On this reading the invariance programme is not superseded and not corrected. It is localized. Its finiteness underwriting is exactly as strong as it ever was on the register where a domain group acts, and the results from Mautner to Bonnay stand untouched. What the present analysis adds is the boundary of that register, drawn from the inside by a hypothesis check rather than from the outside by a competing criterion. The programme succeeds where a group is available and is silent where one is not, and this paper is a description of the silence rather than an argument against the speech.

Falsifiable commitments

A negative result in foundations does not predict an observation, and a paper of this kind that borrows the apparatus of empirical prediction without its substance would be worse than one that states plainly what would refute it. What follows are four commitments, each with an exact refuting condition, each checkable by parties sharing nothing with the author. All four are decidable by construction or by finite computation.

Commitment one, the spectral floor

The commitment. For every real three-by-three matrix A and $D = \text{diag}(1, 1, 0)$, the spectral-norm distance $\|AD - I\|_2$ is at least 1, with equality attained, for instance at $A = I$. More generally, for any rank-deficient operator D on a finite-dimensional space, no left inverse exists and the deficiency is bounded below by 1 in spectral norm. The linear presentation is a convenience and not a requirement: the underlying obstruction is set-theoretic, since a map that is not injective has no left inverse in any category, and the spectral bound merely measures how badly the failure fails in the normed setting.

Method of confirmation. Direct computation. Construct A by any method, form $AD - I$, take the largest singular value. The general claim follows from the kernel argument given under the proposed result and is machine-checkable in a proof assistant in a few lines.

Expected outcome. Every computed value at or above 1.0 to within the arithmetic's precision, exactly 1 on the characterized equality set and strictly above it off that set. Four exhibited minimizers, $A = I$, $A = D$, $A = D^*$, and $A = 0$, each return 1.000000000000.

Null hypothesis. A single exhibited A with $\|AD - I\|_2 < 1$ to a margin exceeding the arithmetic's error bar refutes the obstruction and with it the paper's central mechanism. No further argument would be needed and none would be offered.

Commitment two, the invariant ring is the constants

The commitment. For a register in which every constituent is abstractable, the ring of functions invariant under the three coordinate abstractions is $\mathbb{R}$. Closure under composition is entailed by the invariance relation and is not assumed. Consequently finite generation holds trivially, the ring being generated by the constant one, and separation degenerates rather than fails to exist. Since no invariant distinguishes any pair of points, the separating condition quantifies over an empty set of pairs and is satisfied vacuously, so every

subset of the invariant ring is a separating set, the empty set included. The Derksen-Kemper guarantee is met and is empty. What collapses is not the existence of a separating set but its separating power, which is nil. No transport of the separating-set theorem is required, since the question it would answer is settled directly and in the degenerate direction.

Method of confirmation. Two steps, both decidable. Apply invariance under D_1, D_2, D_3 at the three points $x, (0, x_2, x_3)$, and $(0, 0, x_3)$, which gives $f(x) = f(0, 0, 0)$ in three substitutions. Independently, compute the orbit count on the finite model: one class under the three generators. The semigroup enumeration and the annihilator are a corollary and not a step.

Expected outcome. One orbit class on the finite model; seven elements in the generated semigroup, identity absent, annihilator present; invariant ring the constants. All are exact and none is sampled.

Null hypothesis. Exhibition of a non-constant function invariant under the three coordinate abstractions would refute the collapse outright. Alternatively, an argument that some constituent is not abstractable, so that a generator is absent from the register, would restore a non-trivial invariant ring in the coordinates no admitted abstraction touches and confine the result to the failure of group-hood alone. That is the live route and the one we invite, since it attacks a hypothesis rather than an identity. An argument attacking closure under composition is not a route at all, since closure is entailed by invariance and assuming it away changes nothing. We note that this commitment was open in an earlier draft, pending a separating-set theorem for monoid actions; the collapse makes that theorem unnecessary and the earlier framing is withdrawn.

Commitment three, the domain group does not transport

The commitment. The permutation group underwriting the Tarski-Sher criterion acts on the domain of discourse and admits no faithful action on the space of decompositions of a claim that carries the abstraction operation, since a faithful action would supply inverses to a non-invertible generator.

Method of confirmation. Formal, by unitality. An action of a group G on a set is by definition unital, so the identity of G acts as the identity transformation. If φ is the induced homomorphism into the transformation monoid, then $\varphi(g)\varphi(g^{-1}) = \varphi(e) = \text{id}$ for every g , so every $\varphi(g)$ is invertible in the monoid; D is not invertible, so D is not in the image. Unitality has to be stated rather than assumed, and the idempotent case shows why: the constant map carrying every element of a group to D is a semigroup homomorphism, and its image, the single element D , is a group whose identity is D itself, since $DD = D$. Group-hood of the image is therefore not by itself the contradiction. Invertibility relative to the ambient identity is.

Expected outcome. Every proposed transport either omits abstraction from the image, in which case it acts on a proper invertible

sub-part and does not address the register, or contains it, in which case it is inconsistent.

Null hypothesis. A construction placing abstraction inside a group action on a suitably enlarged space, where the enlargement is not a retreat to the invertible sub-part and not a labelling trick that records the discarded constituent alongside the image, would refute this commitment. We note the obvious near-miss and exclude it explicitly: a map that carries a claim to the pair consisting of its abstraction and its discarded remainder is invertible, but it is not abstraction, since it discards nothing, and a register in which nothing is discarded is not the register under analysis.

Commitment four, the residence of non-effectiveness

The commitment. In any pipeline composing a semantic decomposition, a finite screening roster, and a fixed arithmetic verdict, no reduction of the later two stages can absorb the pipeline's non-effectiveness, because those stages are decidable by construction.

Method of confirmation. Structural. Exhibit the screening roster as a finite enumeration and the verdict stage as a terminating computation over finitely representable inputs, then apply Church's theorem and Tarski's undefinability to the composite^{21,24}.

Expected outcome. Every attempt to mechanize the pipeline end to end either leaves the decomposition stage as an external input, in which case the placement holds, or claims to mechanize it, in which case the composite becomes an effective truth-assignment procedure over an arithmetic domain and is refuted by the classical results.

Null hypothesis. An implemented pipeline that mechanizes the decomposition stage with decidable membership, screens through a finite roster, computes a fixed arithmetic verdict, and demonstrably tracks arithmetic truth on a class where truth is undecidable, would refute this commitment and, more importantly, would refute Church and Tarski, which is why we regard this as the most secure of the four.

On the honest strength of a negative result

The four commitments are not of equal grade and presenting them as if they were would be the inflation this paper's own method is built to catch. Commitments one and four are theorem-grade, the first by elementary linear algebra and the second by the classical undecidability results, and both are refutable only by refuting mathematics that has stood for ninety years. Commitment three is structural and rests on an argument we regard as sound but which is stated rather than formalized. Commitment two is theorem-grade by three substitutions and a finite orbit count, and it was open in an earlier draft: the paper then claimed only that the resident machinery could not run and declined to claim that no closure existed. The three substitutions settle it, and the earlier caution is recorded rather than erased, because the distinction it drew remains the right one wherever a computation is not available. What is conditional is not the algebra, and not closure, which is entailed, but admissibility, and that is where the result should be attacked.

Discussion and implications

The programme gains a boundary rather than a rival. The invariance criterion has been criticized for overgenerating and for undergenerating, and both critiques operate inside the register, arguing that the chosen group admits too much or too little^{6,8,10}. The present analysis is of a different kind and is compatible with either outcome of that debate. It says that whichever group is chosen, the criterion's transportability is bounded by the availability of a group, and identifies the first register one meets where none is available. A programme with a mapped boundary is in better condition than one whose scope is assumed, and nothing above weakens a single result inside the boundary.

A reading of the formalization gap. Practitioners of formal verification and of formal semantics have long observed that writing the specification is harder than discharging it, and the observation is usually offered as a fact about effort. The residence argument recasts it as a fact about structure. If the discharge stages are decidable, the difficulty cannot be there, and no improvement to them relocates it. This predicts a phenomenon that is already observed and rarely explained: automation of the later stages advances rapidly while the specification step resists, not because it has been attacked less but because it is carrying a load the other stages are structurally unable to hold.

Consequences for machine architectures. The geometric deep learning blueprint derives architecture from symmetry²⁹, and its success where a group is identified is not in question. The present result suggests a principled account of where the blueprint's leverage should be expected to thin: not at any particular modality, but wherever the operative transformation is lossy rather than invertible. Abstraction, summarization, and generalization are lossy by definition, and a symmetry-derived inductive bias has correspondingly less to grip. This is a structural expectation rather than an empirical claim, and it is stated as such.

What the result does not license. Three inferences are barred and are named so they cannot be drawn from the paper by omission. It does not follow that the decomposition register is unbounded or unconstrained; the constituent count is fixed by the decomposition discipline and by an algebraic argument that has nothing to do with invariance, and losing a closure theorem does not gain a degree of freedom. It does not follow that a natural-language register is superior to a formal one; a faculty with no terminal test is a faculty that never concedes, and an unbounded capacity for re-description is an unbounded capacity for motivated re-description, which is a liability and not a licence. And it does not follow that mathematics is a language in any sense that would collapse the registers, since a numerical verdict is invariant under re-description of the claim it verifies and that invariance is exactly the contribution the arithmetic stage makes.

Anticipated objection, the register is an artefact. One might hold that the decomposition register is an artefact of a particular verification discipline and that a different discipline would not raise

the question. The reply is that any procedure which isolates a constituent of a claim in order to examine it separately performs an abstraction, and any abstraction is a lossy map. The register is not proprietary. It is the register of every method that carves before it computes, which is every method that computes on structured claims at all.

Anticipated objection, the operator is a toy. The matrix D is small and simple, and one might read the argument as depending on a caricature. The reply is that the smallness is the point. The obstruction is not delicate and does not require an elaborate model to appear; any rank-deficient linear map exhibits it, and the general statement in the first commitment quantifies over all of them. A caricature that suffices to violate a hypothesis is enough to block the theorem that hypothesis serves.

Limitations. Four are load-bearing. Admissibility, that every constituent is abstractable, is the load-bearing premise and is the most likely route to refutation, since granting it settles the invariant ring outright and denying it restores a non-trivial one; the appendix supplies it for the counting operation and its extension to ordinary use is the honest gap. The residence argument is a placement supported by a decidability observation, and while we regard it as secure we have not formalized the pipeline in a proof assistant. The identification of abstraction as the operative transformation on the decomposition register is an interpretive step: it is well motivated and, we believe, forced by what decomposition is, but it is not a theorem and is not presented as one. And the fourth, carried openly: the collapse is stated for sharp coordinate abstractions, and a register admitting graded or partial ones has a different transformation set. We compute that case rather than leave it, at LOG-CHK.9, and the collapse survives it for continuous invariants and on any finite model, the escape requiring invariants discontinuous along attenuation orbits; what we have not computed is a register mixing sharp and graded abstractions on different constituents.

Conclusion

The invariance programme in logic has run for eighty years on a register where a group acts by construction, and the finiteness theorem behind it has therefore never been asked to do anything its hypotheses do not cover. This paper asks it, one register over, and reports that it cannot answer. On the register where a claim is decomposed rather than interpreted, the operative transformation is abstraction, which is idempotent and has no left inverse, with the deficiency bounded below by exactly one in spectral norm; the transformation set is a monoid; group-hood fails before reductivity is tested; and neither Hilbert's finiteness theorem, nor Weyl's explicit generators, nor the separating-set results reachable from them, apply by transport. Admissibility of every constituent then forces every invariant to be constant, in three substitutions and with closure entailed rather than assumed, so the invariant ring is the reals and the criterion at this register demarcates nothing.

The primary refuting condition is exact and cheap to check: a single matrix A with $\|AD - I\|_2$ below 1 by more than the arithmetic's error

bar ends the argument. The most likely genuine refutation is not the spectral floor but admissibility: an argument that some constituent of a claim is not abstractable in the register, so that a generator is absent, would restore a non-trivial invariant ring in the coordinates no admitted abstraction touches and confine the result to the failure of group-hood alone. We invite that argument specifically, since it attacks a premise rather than an identity, and the object it would name, the register in which only some constituents are abstractable, has an invariant ring we compute here for the two- and one-generator cases and leave open for no sharp case.

We ask two things of the relevant communities. From the invariance-criteria literature, a ruling on whether the boundary drawn here is the right one, and if not, an identification of the acting group we have failed to find. From semigroup and invariant theory, a ruling on whether the degenerate case identified here is the general shape of invariant theory under a transformation semigroup with a zero, or an artefact of this particular family, which is a well-posed algebraic question independent of every philosophical claim in this paper.

The open question we regard as most important is not whether the invariant theory collapses but what follows from its collapsing: if the non-effectiveness of a verification pipeline is structurally confined to its decomposition stage, then the discipline governing that stage is not an approximation awaiting mechanization but the load-bearing component, and the standard practice of treating it as a temporary limitation has the typing backwards.

Accepting the result would mean reading the invariance criterion as a local instrument with a mapped boundary rather than a general method awaiting extension, and reading the informal step at the head of every formal pipeline not as the part not yet automated but as the part that provably cannot be, on pain of the pipeline deciding what is undecidable.

Appendix A: Foundational principles, stated independently

The following principles are drawn from a broader epistemic framework and are presented here as standalone mathematical or physical statements, each independently motivated and independently testable within the native discipline. None is required to accept the argument above, which rests on the cited classical results.

The three-constituent decomposition. A complete factual claim decomposes into exactly three mutually irreducible constituents: what is asserted to be the case, the dynamical or kinetic content of that assertion, and the registrational content, the fact of its being put forward from a standpoint as holding. The count is established operationally by a deletion discipline, deleting each constituent in turn and observing that the residue is no longer a complete claim, and independently by an algebraic argument: an audit composition law requiring associativity and the absence of zero divisors admits, among the finite-dimensional associative division algebras over the reals, only the reals, the complex numbers, and the quaternions by Frobenius²⁶, the finite-dimensionality being a hypothesis of that theorem and not an omission. Three mutually orthogonal imaginary axes require

dimension at least four, and among the finite-dimensional real normed division algebras only the quaternions carry exactly three; the octonions carry seven and fail associativity, so the construction stops at three.

Independence before quantification. The three constituents admit independent variation: varying the subject with predicate and assertoric mode fixed moves the first alone, and correspondingly for the others. This is an orthogonality read off the structure of the claim rather than imposed by a coordinate system, and it is what licenses treating the constituents as axes at all.

The verdict as a determinant. Where the three constituents are quantified as rows over a common set of evaluation contexts, normalized, and read as unit vectors, their independence is measured by the determinant of their correlation matrix, which lies in the closed unit interval by Hadamard's inequality above and positive-semidefiniteness below. A determinant clear of zero is a genuine three-dimensional configuration; a collapse to zero is one row lying in the plane of the other two. The determinant equals the square of the signed volume, where the volume is computed with the three rows expressed in an orthonormal basis of the subspace they span, and is therefore invariant under reflection of any row, so it certifies dimensionality and never direction, and any directional reading must come from the ordered constituents rather than from the scalar.

Warrant typing. Every claim carries an explicit grade, theorem, conditional, structural, or premise, and no claim is stated above its grade. A composite claim inherits the grade of its weakest load-bearing component. This is a discipline rather than a theorem and is stated here because it governs how the four commitments are to be read.

Appendix B: Adjacent problem structures

Two structures in adjacent literatures share the shape of the result and are recorded as connections rather than as claims.

Lossy compression and the non-existence of an inverse. The obstruction is formally the statement that a rank-deficient map has no left inverse, which is also the statement that lossy compression is not invertible. The connection is not decorative: an abstraction is a compression that selects which coordinates survive, and the literature on rate-distortion has a developed account of what is recoverable and at what cost. What that literature does not ask, because its objects are signals rather than claims, is whether the compression maps form a group, and the answer there is the same as here and for the same reason.

Diagonal arguments and the confession of a ladder. Lawvere showed that Cantor's theorem, Russell's paradox, Gödel's first incompleteness theorem, and Tarski's undefinability theorem are instances of one fixed-point scheme in a cartesian closed category³³, a unification later extended by Yanofsky³⁴. The residence argument uses Tarski and Church as external constraints, and the Lawvere reading suggests they are one constraint seen twice. Whether the decomposition register's failure to admit a group is related to that

scheme, or is an independent obstruction that merely co-occurs with it, we do not know, and we record the question without a conjecture.

Author's provenance and method disclosure

This paper was developed under Trisduction, a verification and organizing discipline, not a source of results. Its conclusions rest solely on the standard results cited in the body, which the method was used to decompose, assemble, and audit; the method adds them no warrant and claims no authorship over them. The full method statement is at the master reference in the References below³⁸.

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Data and code availability

The battery LOG-CHK is nine short computations: two sampled checks at seed 20260622 and seven exact ones requiring no seed, over three-by-three matrices, a finite closure enumeration, a twenty-seven-point orbit count, and a sixty-four-point graded model with its continuity argument, all reproducible from the specification under the proposed result and in Box 1 in any standard numerical stack. No dataset is required.

Competing interests

The author declares no competing financial interests. The method disclosure above states the non-financial provenance interest in full.

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The prior-art position of this paper was established by adversarial audit against the invariance-criteria and invariant-theory literatures, and an earlier framing that claimed novelty for the transport of the Erlangen criterion to a logical register was withdrawn on discovery of Mautner 1946. The correction is recorded here rather than concealed.